Ex-Ante Interaction Regressions

Type 0 vs Type 1 | Oreopoulos replication extension

Contents

1 Research objective and specification **1**

2 Setup and data preparation **2**

3 Baseline average treatment effect **3**

4 Three ex-ante interaction regressions **4**

5 Regression-implied subgroup treatment effects **7**

6 Interpretation and comparison notes **8**

7 Optional: joint interaction specification **8**

1 Research objective and specification

We estimate the immigrant-sounding name penalty in employer callback rates and test whether it differs by three **pre-specified moderators**: language skills, extracurricular activities, and gender. The regression estimates provide a theory-driven counterpart to the causal forest's subgroup treatment-effect estimates; this is **not** a predictive-performance horse race.

Moderator (X)	Variable	Question	Expected interaction
Language skills	`language_skills`	Does listing language proficiency reduce the foreign-name callback penalty?	Positive
Extracurricular activities	`extracurricular_skills`	Do extracurricular signals reduce the foreign-name callback penalty?	Positive
Gender	`female`	Is the foreign-name callback penalty less negative for female than male applicants?	Positive

The positive signs are **ex-ante hypotheses**, not assumed findings. Gender is motivated by gender-differentiated ethnic discrimination, not necessarily by a productivity-signal mechanism.

2 Setup and data preparation

The analysis uses the same cleaned Type 0/1 sample (`df_int1`) as the balance-table and causal-forest scripts. Treatment $D = 1$ denotes a foreign-sounding name (Type 1), and $D = 0$ an English-sounding name (Type 0). Outcome Y is the callback indicator. Inference is clustered by `firmid`.

```
required <- c("dplyr", "estimatr", "knitr")
missing <- setdiff(required, rownames(installed.packages()))
if (length(missing)) install.packages(missing, repos="https://cloud.r-project.org")
library(dplyr)
library(estimatr)
library(knitr)
```

```
if (!file.exists("cleaned data.Rda")) stop("Run cleaned_data.Rmd first, or put cleaned data.Rda in this
loaded <- load("cleaned data.Rda")
stopifnot("df_int1" %in% loaded)
required_cols <- c("D", "callback", "firmid", "fall_data", "type", "language_skills", "extracurricular_s
stopifnot(all(required_cols %in% names(df_int1)))
dat <- df_int1 %>% mutate(
  D = as.numeric(as.character(D)),
  callback = as.numeric(as.character(callback)),
  wave = factor(fall_data)
)
stopifnot(all(na.omit(dat$D) %in% 0:1), all(na.omit(dat$callback) %in% 0:1))
cat("Observations:", nrow(dat), "\n")
```

```
## Observations: 5657
```

```
print(table(dat$D, useNA="ifany"))
```

```
##
##    0    1
## 3026 2631
```

```
print(table(dat$type, dat$D, useNA="ifany"))
```

```
##
##        0    1
##   0 3026    0
##   1    0 2631
```

```
cat("Firms:", dplyr::n_distinct(dat$firmid), "\n")
```

```
## Firms: 3269
```

`same_exp` is excluded from the primary controls because it is systematically missing in wave 2; including it would remove that wave. The common main-effect control set is `language_skills`, `extracurricular_skills`, `female`, `ba_quality`, `ma`, and `exp_highquality`.

3 Baseline average treatment effect

We estimate a wave-adjusted linear probability model:

$$Y_i = \beta_0 + \beta_1 D_i + \delta_{w(i)} + \varepsilon_i.$$

Here β_1 is the wave-adjusted name effect, with firm-clustered CR2 standard errors. We also report a covariate-adjusted version as a robustness specification.

```
ate_wave <- lm_robust(callback ~ D + wave, data=dat, clusters=firmid, se_type="CR2")
ate_adjusted <- lm_robust(callback ~ D + language_skills + extracurricular_skills + female +
                            ba_quality + ma + exp_highquality + wave,
                          data=dat, clusters=firmid, se_type="CR2")


extract_D <- function(model, label) {

  results <- broom::tidy(model, conf.int = TRUE)

  treatment <- results[results$term == "D", ]

  if (nrow(treatment) != 1) {
    stop("Could not find treatment coefficient D")
  }

  data.frame(
    model = label,
    estimate = treatment$estimate,
    se = treatment$std.error,
    ci_low = treatment$conf.low,
    ci_high = treatment$conf.high,
    p_value = treatment$p.value,
    n = nobs(model)
  )
}



ate_table <- bind_rows(
  extract_D(ate_wave, "Wave-adjusted"),
  extract_D(ate_adjusted, "Wave + covariate adjusted")
) %>%
  mutate(
    p_value = ifelse(
      p_value < 0.001,
      "<0.001",
      sprintf("%.4f", p_value)
    )
  )

kable(
  ate_table,
  digits = 4,
```

```
  caption = "Average name effect (foreign minus English name)"
)
```

Table 1: Average name effect (foreign minus English name)

model	estimate	se	ci_low	ci_high	p_value	n
Wave-adjusted	-0.0443	0.0069	-0.0579	-0.0307	<0.001	5657
Wave + covariate adjusted	-0.0443	0.0069	-0.0578	-0.0307	<0.001	5657

Note: The standard error (SE) reported above differs from that in Oreopoulos (2011) due to differences in the estimation method. Oreopoulos uses heteroskedasticity-robust (HC1) standard errors, whereas our analysis uses firm-clustered CR2 standard errors (SE = 0.00693) to account for potential within-firm dependence, since multiple resumes were submitted to the same employer. Both methods produce the same ATE estimate (-0.04434), with the difference in SE arising from the variance estimation method. The table below compares the results using HC1 and CR2, showing that HC1 reproduces Oreopoulos's reported SE of 0.009.

```
#replicating exact Oreopoulos results using HC1 SE

model_HC1 <- lm_robust(
  callback ~ D + wave,
  data = dat,
  se_type = "HC1"
)

#the CR2 SE that we are using instead

model_CR2 <- lm_robust(
  callback ~ D + wave,
  data = dat,
  clusters = firmid,
  se_type = "CR2"
)

# Compare treatment estimates and standard errors
data.frame(
  method = c("HC1", "CR2"),
  estimate = c(coef(model_HC1)["D"],
               coef(model_CR2)["D"]),
  SE = c(model_HC1$std.error["D"],
         model_CR2$std.error["D"])
)
```

```
##   method    estimate          SE
## 1    HC1 -0.04434211 0.008796044
## 2    CR2 -0.04434211 0.006933685
```

4 Three ex-ante interaction regressions

For each moderator X, we estimate:

$$Y_i = \beta_0 + \beta_1 D_i + \beta_2 X_i + \beta_3 (D_i \times X_i) + \boldsymbol{\gamma}' Z_i + \delta_{w(i)} + \varepsilon_i.$$

Z contains the **other five** common controls (the focal moderator is already included as a main effect). β_3 tests whether the name effect differs between $X = 1$ and $X = 0$. The models are fitted separately, using the same full analysis sample where complete.

```
moderators <- c("language_skills", "extracurricular_skills", "female")
controls <- c("language_skills", "extracurricular_skills", "female",
              "ba_quality", "ma", "exp_highquality")
fit_interaction <- function(x) {
  rhs <- c("D", x, paste0("D:", x), setdiff(controls, x), "wave")
  f <- reformulate(rhs, response="callback")
  lm_robust(f, data=dat, clusters=firmid, se_type="CR2")
}
models <- setNames(lapply(moderators, fit_interaction), moderators)

interaction_table <- bind_rows(lapply(moderators, function(x) {

  m <- models[[x]]
  term <- paste0("D:", x)

  # Extract regression results with confidence intervals
  results <- broom::tidy(m, conf.int = TRUE)

  # Select the interaction coefficient
  interaction <- results[results$term == term, ]

  if (nrow(interaction) != 1) {
    stop(paste("Interaction term not found:", term))
  }

  data.frame(
    moderator = x,
    interaction = interaction$estimate,
    se = interaction$std.error,
    p_value = interaction$p.value,
    ci_low = interaction$conf.low,
    ci_high = interaction$conf.high,
    n = nobs(m)
  )
}))

kable(interaction_table, digits=4, caption="Ex-ante interaction coefficients (D x X)")
```

Table 2: Ex-ante interaction coefficients (D x X)

moderator	interaction	se	p_value	ci_low	ci_high	n
language_skills	-0.0144	0.0200	0.4715	-0.0535	0.0248	5657
extracurricular_skills	0.0307	0.0183	0.0939	-0.0052	0.0667	5657
female	-0.0366	0.0171	0.0323	-0.0701	-0.0031	5657

Note: The estimates from our interaction regressions differ from those reported in Oreopoulos (2011, Table 5 Panel B) because they estimate different quantities using different regression specifications.

Oreopoulos (2011, Table 5 Panel B) reports the effects of resume characteristics on callback rates from separate regressions for each resume type (effect of X for Type 0 and effect of X for Type 1). In contrast, our analysis pools Type 0 and Type 1 resumes and directly estimates the interaction between name treatment and each ex-ante moderator. This interaction represents the *difference* between the effects of X for Type 1 and Type 0 (effect of X for Type 1 - effect of X for Type 0), allowing us to directly test whether the immigrant-name penalty varies across each moderator.

To verify this distinction, we replicate Oreopoulos's separate regressions for Types 0 and 1 and compare their coefficients with our pooled interaction estimates in the table below. The replication reproduces Oreopoulos's published coefficients to three decimal places.

Our interaction coefficients are close to, but not identical to, the differences implied by Oreopoulos's results because the regression specifications differ. **Oreopoulos's separate regressions allow all covariate and wave effects to vary by resume type, whereas our pooled regressions interact only the selected moderator with treatment, holding the effects of other controls and wave constant across types.** For example, the difference in the extracurricular activities coefficients (0.023 - (-0.007) = 0.030) is close to our estimated interaction coefficient of 0.0307.

```
# To replicate Oreopoulos's result, try to run separate regressions for Type 0 and Type 1.

# Prepare a separate dataset for replication; deal with missing same_exp values
dat_oreo <- dat %>%
  mutate(same_exp = ifelse(is.na(same_exp), 0, same_exp))

# Type 0: English-sounding names; note how the data is filtered to only have the samples where D == 0
oreo_type0 <- lm_robust(
  callback ~ female + ba_quality + extracurricular_skills +
    language_skills + ma + same_exp + exp_highquality + wave,
  data = filter(dat_oreo, D == 0),
  se_type = "HC1"
)

# Type 1: Foreign-sounding names; note how the data is filtered to only have the samples where D == 1
oreo_type1 <- lm_robust(
  callback ~ female + ba_quality + extracurricular_skills +
    language_skills + ma + same_exp + exp_highquality + wave,
  data = filter(dat_oreo, D == 1),
  se_type = "HC1"
)

# Pull coefficients into table for comparison

oreo_comparison <- bind_rows(lapply(moderators, function(x) {

  b0 <- unname(coef(oreo_type0)[x])
  b1 <- unname(coef(oreo_type1)[x])

  data.frame(
    moderator = x,
    oreo_type0 = b0,
    oreo_type1 = b1,
    difference = b1 - b0,
    our_interaction = interaction_table$interaction[
```

```
      interaction_table$moderator == x
    ]
  )
}))

knitr::kable(
  oreo_comparison,
  digits = 3,
  col.names = c(
    "Moderator",
    "Oreopoulos Type 0",
    "Oreopoulos Type 1",
    "Type 1 - Type 0",
    "Our interaction"
  ),
  caption = "Comparison with Oreopoulos (2011), Table 5 Panel B"
)
```

Table 3: Comparison with Oreopoulos (2011), Table 5 Panel B

Moderator	Oreopoulos Type 0	Oreopoulos Type 1	Type 1 - Type 0	Our interaction
language_skills	0.020	0.007	-0.012	-0.014
extracurricular_skills	-0.007	0.023	0.030	0.031
female	0.049	0.012	-0.037	-0.037

5 Regression-implied subgroup treatment effects

For each binary moderator, the model-implied treatment effects are:

$$\widehat{\tau}(X = 0) = \widehat{\beta}_1, \qquad \widehat{\tau}(X = 1) = \widehat{\beta}_1 + \widehat{\beta}_3.$$

For $X = 1$, the variance must include the covariance between the two coefficient estimates:

$$\widehat{\mathrm{Var}}(\widehat{\beta}_1 + \widehat{\beta}_3) = \widehat{\mathrm{Var}}(\widehat{\beta}_1) + \widehat{\mathrm{Var}}(\widehat{\beta}_3) + 2\widehat{\mathrm{Cov}}(\widehat{\beta}_1, \widehat{\beta}_3).$$

The following code uses each model's firm-clustered covariance matrix and its degrees of freedom to calculate 95% confidence intervals. These subgroup effects, not the raw interaction coefficients, will be compared with `average_treatment_effect(cf, subset = ...)` from the causal forest.

```
subgroup_effects <- bind_rows(lapply(moderators, function(x) {
  m <- models[[x]]
  b <- coef(m)
  V <- vcov(m)
  interaction <- paste0("D:", x)
  out <- lapply(0:1, function(z) {
    a <- setNames(rep(0, length(b)), names(b))
    a["D"] <- 1
    if (z == 1) a[interaction] <- 1
    estimate <- sum(a * b)
```

```
    se <- sqrt(drop(t(a) %*% V %*% a))
    # estimatr stores coefficient-specific degrees of freedom
    df <- min(m$df[match(c("D", interaction), names(b))], na.rm=TRUE)
    crit <- qt(0.975, df=df)
    data.frame(moderator=x, subgroup=paste0("X=", z),
               estimate=estimate, se=se,
               ci_low=estimate-crit*se, ci_high=estimate+crit*se,
               n_subgroup=sum(dat[[x]] == z, na.rm=TRUE))
  })
  bind_rows(out)
}))
kable(subgroup_effects, digits=4,
      caption="Regression-implied subgroup name effects and 95% confidence intervals")
```

Table 4: Regression-implied subgroup name effects and 95% confidence intervals

moderator	subgroup	estimate	se	ci_low	ci_high	n_subgroup
language_skills	X=0	-0.0406	0.0084	-0.0570	-0.0242	4233
language_skills	X=1	-0.0550	0.0167	-0.0878	-0.0221	1424
extracurricular_skills	X=0	-0.0628	0.0130	-0.0883	-0.0372	2240
extracurricular_skills	X=1	-0.0320	0.0101	-0.0518	-0.0123	3417
female	X=0	-0.0257	0.0107	-0.0466	-0.0048	2781
female	X=1	-0.0623	0.0113	-0.0844	-0.0401	2876

6 Interpretation and comparison notes

- A negative effect denotes a callback penalty for foreign-sounding names.
- A positive interaction means the name penalty is **less negative** for $X = 1$ than for $X = 0$.
- The three interaction regressions are separate pre-specified tests; the estimated subgroup effects are model-implied and conditional on the included main-effect controls and wave.
- The causal forest comparison should report doubly robust subgroup ATEs with firm clustering, side by side with these regression estimates and confidence intervals. Neither method observes the true individual CATE.
- Statistical significance of an interaction is not proof of a particular discrimination mechanism. In particular, gender differences need not arise from statistical discrimination.

7 Optional: joint interaction specification

A joint model can be considered as a supplementary analysis, but is not required for the three primary ex-ante tests.

```
joint <- lm_robust(
  callback ~ D * (language_skills + extracurricular_skills + female) +
    ba_quality + ma + exp_highquality + wave,
  data=dat, clusters=firmid, se_type="CR2"
)
summary(joint)
```